

gradient

the gradient describes how something changes specifically, how a function changes as you move in different directions

if you have $f(x,y)$ the gradient of f is a vector that points in the direction of the steepest increase of the function which is written as:

by steep we mean something rising or falling sharply nearly perpendicular

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \text{ or } \nabla f = (f_x, f_y)$$

this just means the partial derivative of x and partial derivative of y

if you have more variables, like $f(x,y,z)$ then:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \text{ or } \nabla f = (f_x, f_y, f_z)$$

Concept	Meaning
Gradient symbol	∇f
Type	Vector
Direction	Steepest increase
Magnitude	Rate of steepest increase
Negative gradient	Steepest decrease

- directional derivative

if we have a function $f(x, y)$ (think of this as a hill surface)

a point $P = (x_0, y_0)$ (think of this as where you are standing)

a direction vector $u = (u_1, u_2)$ (the direction you want to walk)

★ the direction vector u is a unit vector

• review on unit vector

$$V = \frac{V}{|V|}$$

Vector
magnitude
of vector

the directional derivative of f at point P in the direction of u tells you how fast f changes if you move in that direction

$$D_u f = \vec{\nabla} f \cdot \vec{u} \quad @ \quad P$$

dot product

So:

the gradient tells you the steepest direction and steepness overall

the directional derivative tells you how steep it is in the specific direction you choose

$$f(x, y) = xy^3 - x^2, (1, 2), \theta = \frac{\pi}{3}$$

$$\text{Unit Vector use} = \cos \theta \hat{i} + \sin \theta \hat{j}$$

$$u = \cos(\pi/3) \hat{i} + \sin(\pi/3) \hat{j}$$

$$u = \frac{1}{2} \hat{i} + \frac{\sqrt{3}}{2} \hat{j}$$

$$\vec{\nabla} f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \Rightarrow \langle y^3 - 2x, 3y^2x \rangle @ (1, 2) = \langle 6, 12 \rangle$$

$$\frac{\partial f}{\partial x} = y^3 - 2x$$

$$D_u f = \vec{\nabla} f \cdot \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$$

$$(6)\left(\frac{1}{2}\right) + (12)\left(\frac{\sqrt{3}}{2}\right) = \boxed{2 + 6\sqrt{3}}$$

$$\frac{\partial f}{\partial y} = 3y^2x$$

$$f(x, y) = x^2 \ln(y), P(3, 1), \vec{u} = -\frac{5}{13} \hat{i} + \frac{12}{13} \hat{j}$$

$$\frac{\partial f}{\partial x} = 2x \ln(y)$$

$$\frac{\partial f}{\partial y} = \frac{x^2}{y}$$

$$\vec{\nabla} f = \left\langle 2x \ln(y), \frac{x^2}{y} \right\rangle @ P(3, 1) = \left\langle 0, \frac{1}{3} \right\rangle$$

$$D_u f = \vec{\nabla} f \cdot \left\langle 0, \frac{1}{3} \right\rangle \cdot \vec{u} \left\langle -\frac{5}{13}, \frac{12}{13} \right\rangle$$

$$(0)\left(-\frac{5}{13}\right) + \left(\frac{1}{3}\right)\left(\frac{12}{13}\right) = \boxed{\frac{4}{13}}$$

$$g(s, t) = s^2 e^{-t}, \quad (3, 0), \quad \vec{V} = 3\hat{i} + 4\hat{j}$$

$$g_s = 2s e^{-t}$$

$$g_t = -e^{-t} s^2$$

$$\vec{\nabla} f = \langle 2s e^{-t}, -e^{-t} s^2 \rangle \quad @ \quad P(3, 0) = \langle 6, -9 \rangle$$

You were given the vector not the unit vector you need to get the unit vector

$$V = \langle 3, 4 \rangle = \sqrt{(3)^2 + (4)^2} = 5$$

$$\vec{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$$

$$D_{\vec{u}} f = \vec{\nabla} f \cdot \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$$

$$(6) \left(\frac{3}{5} \right) + (-9) \left(\frac{4}{5} \right) = \boxed{\frac{-18}{5}}$$

$$f(x, y) = \frac{x}{y^2}, \quad P(3, -1), \quad Q(-2, 11)$$

* Notice here you were given two points to find the unit vector you need to $\vec{PQ}$
 $(Q - P)$ then take the magnitude and divide by it to get the unit vector

$$Q(-2, 11) - P(3, -1) \Rightarrow \langle -5, 12 \rangle$$

$$\sqrt{(-5)^2 + (12)^2} = 13$$

$$\left\langle -\frac{5}{13}, \frac{12}{13} \right\rangle$$

$$\vec{\nabla} f \cdot \left\langle \frac{1}{y^2}, -\frac{2x}{y^3} \right\rangle \Rightarrow \langle 1, -6 \rangle$$

$$D_{\vec{u}} f = \vec{\nabla} f \cdot \vec{u} = \langle 1, -6 \rangle \cdot \left\langle -\frac{5}{13}, \frac{12}{13} \right\rangle$$

$$(1) \left(-\frac{5}{13} \right) + (-6) \left(\frac{12}{13} \right) = \boxed{\frac{-67}{13}}$$

$$F_x = \frac{1}{y^2}$$

$$F_y = -\frac{2x}{y^3}$$

Find the Maximum rate of Change of f at the given Point and direction at which it occur

$$f(x,y) = \frac{x}{x^2+y^2} \text{ at } (-1,1)$$

*You were given the Point but not the unit Vector So you need to Find the unit Vector

$$\vec{\nabla} \cdot f = \langle f_x, f_y \rangle \Rightarrow \left\langle \frac{(x^2+y^2)-2x^2}{(x^2+y^2)^2}, \frac{(x^2+y^2)-2xy}{(x^2+y^2)^2} \right\rangle @ (-1,1)$$

$$\vec{\nabla} \cdot f(-1,1) = (0, \frac{1}{2})$$

★ to Find the unit Vector you use this equation:

$$\vec{u} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|}$$

$$\vec{\nabla} \cdot f(-1,1) = (0, \frac{1}{2})$$

$$|\nabla f| = \sqrt{(0)^2 + (\frac{1}{2})^2} = \frac{1}{2}$$

$$\vec{u} = \frac{(0, \frac{1}{2})}{\frac{1}{2}} = (0,1)$$

$$D_{\vec{u}} f = \vec{\nabla} f(0, \frac{1}{2}) \cdot \vec{u}(0,1)$$

$$(0)(0) + (\frac{1}{2})(1) = \boxed{\frac{1}{2}}$$